Test, at 5% level of significance, whether the difference between the proportions of families in the two localities agreed the statement.

68. One thousand apples kept under one type of storage were found to show rotting to the extent of 4% and 1500 apples kept under another kind of storage showed 3% rotting. Can it be reasonably concluded at 5% level of significance that the second type of storage is superior to the first?

69. Preference of school students, who participate in Sports events, to do physical exercises in Modern Gymnasium rather than doing aerobic exercises was analyzed. The number of students randomly selected from two States and their preference for Modern Gymnasium are given below.

State	No. of Students	
	Sampled	Preferred Modern Gymnasium
A	50	38
B	60	52

Test whether the difference between proportions of school students who prefer Modern Gymnasium to do their exercises in the two States is significant at 5% level of significance.

70. Interest of XII Students on Residential Schooling was investigated among randomly selected students from two regions. Among 300 students selected from Region A, 34 students expressed their interest. Among 200 students selected from Region B, 28 students expressed their interest. Does this information provide sufficient evidence to conclude at 5% level of significance that students in Region A are more interested in Residential Schooling than the students in Region B?

ACTIVITIES

1. In your institution, collect data from your Physical Education department about height/weight of the students in a particular class (Standard 9). Take a random sample of 50 students from your school and find the mean and standard deviation for their height/weight. Verify through the statistical tests of inferential statistics, Find the significant difference between sample mean and population mean. Give your comments.
2. Consider two group of students of same class (Standard 10) taking an examination in a particular subject in Tamil medium and in English medium in your school. Get samples from two medium and find their mean and standard deviation. Is there any significant difference between the performance between the students taking their examination in Tamil medium and English medium? Discuss with your friends.

Note: The teacher and students discuss and they may create their own problems similar to the above problems and expand this exercises.

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